

# Sine Rule, Cosine Rule and Area of a Triangle



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No right angle? Look at what you know. A matching pair (side and its opposite angle) → sine rule.  
Two sides and the angle between, or three sides → cosine rule.

## THE TOPIC ON ONE MAP

### Sine Rule, Cosine Rule and Area of a Triangle

#### 01 Area $\frac{1}{2}ab \sin C$

two sides and the angle between

area given, find a side

#### 02 The sine rule

a side and its opposite angle

the obtuse (ambiguous) case

#### 03 The cosine rule

third side

angle from three sides

#### 04 Multi-step problems

shared side in two triangles

3D flagpole

**WHICH RULE?** Decide from what you are given:

#### ZONE SINE

a matching pair

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

#### ZONE COSINE

two sides + angle  
between, or 3 sides

$$a^2 = b^2 + c^2 - 2bc \cos A$$

#### ZONE AREA

two sides + angle  
between

$$\frac{1}{2}ab \sin C$$

# 01 Area of a Triangle $\frac{1}{2}ab \sin C$

Two sides and the angle **between** them. If the area is given, the same formula finds a missing side or angle.

## WHEN YOU SEE → YOU DO

**two sides and the included angle**

→  $\frac{1}{2}ab \sin C$

**“the area is  $74 \text{ cm}^2$ , find  $EF$ ”**

→ solve  $\frac{1}{2}x^2 \sin 130^\circ = 74$

**a kite or a parallelogram**

→ two equal triangles

**the angle is not between the sides**

→ find it first (sine rule or angle sum)

## WORKED MODEL • JUNE 2022 P2HR

Kite  $ABCD$ :  $AB = 6$ ,  $BC = 11$ , angle  $ABC = 118^\circ$ . Area?

1 triangle  $ABC = \frac{1}{2}(6)(11) \sin 118^\circ = 29.14$

half the kite

**ANSWER**  $2 \times 29.14 = 58.3 \text{ cm}^2$

## IDEA 01

**Practise: Area**  $\frac{1}{2}ab \sin C$ 

My questions first, then selected practice, then real past papers.

**Q1** DR ESLAM Area

CH 4 • TOPIC 11 2 marks

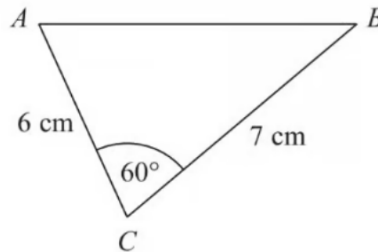
Find the area of a triangle with sides 7 cm and 9 cm and angle  $50^\circ$  between them.

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## SELECTED PRACTICE

**Q2** PRACTICE Practice • Medium

CH 4 • TOPIC 11 5 marks

**1 (a)**  $ABC$  is a triangle.Diagram NOT  
accurately drawnWork out the area of triangle  $ABC$ .

Give your answer correct to 3 significant figures.

(2 marks)

**(b)** Work out the length of the side  $AB$ .

Give your answer correct to 3 significant figures.

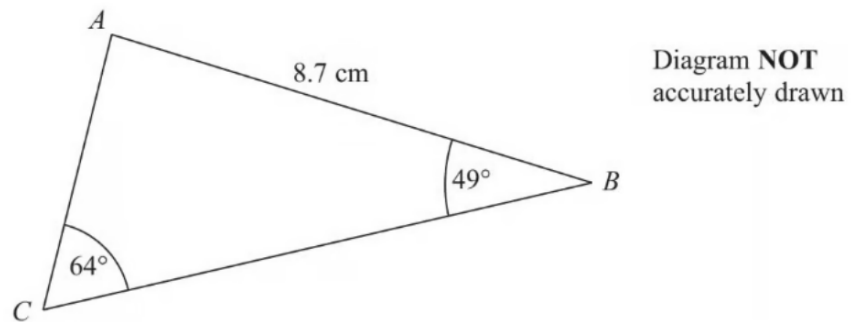
(3 marks)

**Q3****PRACTICE**

Practice • Hard

CH 4 • TOPIC 11 5 marks

2



$ABC$  is a triangle.

$AB = 8.7$  cm.

Angle  $ABC = 49^\circ$ .

Angle  $ACB = 64^\circ$ .

Calculate the area of triangle  $ABC$ . Give your answer correct to 3 significant figures.

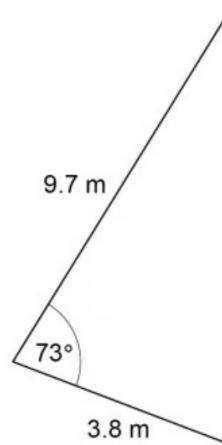
(5 marks)

**Q4****PRACTICE**

Practice • Hard

CH 4 • TOPIC 11 8 marks

12 (a) Here is a triangular sail.



Not drawn  
accurately

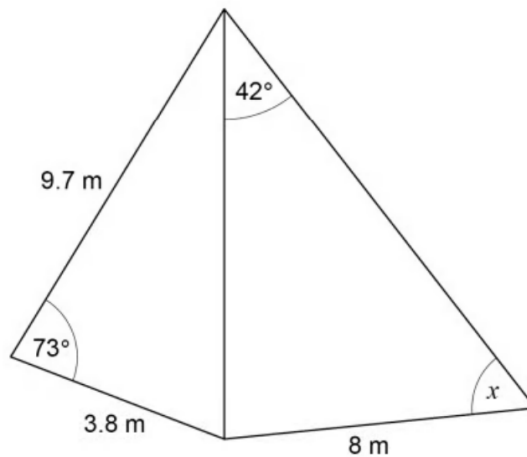
Vicky needs to buy waterproofing liquid for the sail.

She will put **3 coats** of liquid on each side of the sail.  
A litre of liquid covers 8.5 square metres of sail.

How many 1-litre bottles of liquid does Vicky need?

(3 marks)

(b) Another sail is joined to the first sail as shown.



Not drawn  
accurately

$x$  is an acute angle.

Work out the size of angle  $x$ .

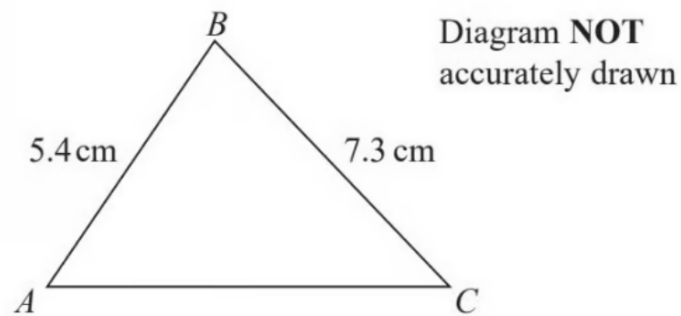
(5 marks)

**Q5****PRACTICE**

Practice • Very Hard

CH 4 • TOPIC 11 6 marks

5  $ABC$  is an acute angled triangle.



The area of triangle  $ABC$  is  $19 \text{ cm}^2$ .

Work out the size of angle  $ACB$ .

Give your answer correct to 3 significant figures.

(6 marks)

## PAST PAPERS

**Q6** PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1H • Q9

CH 4 • TOPIC 11 4 marks

- 9 The diagram shows isosceles triangle  $ABC$

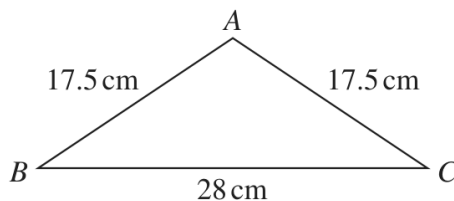


Diagram **NOT**  
accurately drawn

$$AB = AC = 17.5\text{ cm}$$

$$BC = 28\text{ cm}$$

Calculate the area of triangle  $ABC$

.....  $\text{cm}^2$

(Total for Question 9 is 4 marks)

**Q7****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q15

**CH 4 • TOPIC 11 3 marks**

15 The diagram shows a kite  $ABCD$

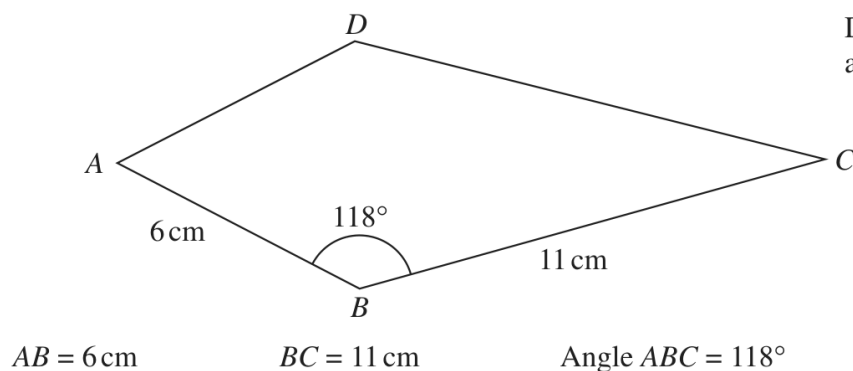


Diagram **NOT**  
accurately drawn

Calculate the area of the kite.

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$

(Total for Question 15 is 3 marks)

**Q8** **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1HR • Q25**CH 4 • TOPIC 11** **6 marks**

25 Here is a triangle  $ABC$

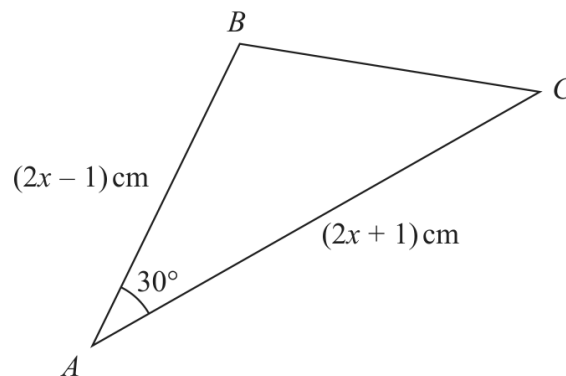


Diagram **NOT**  
accurately drawn

The area of the triangle is  $(x^2 + x - 3.75) \text{ cm}^2$

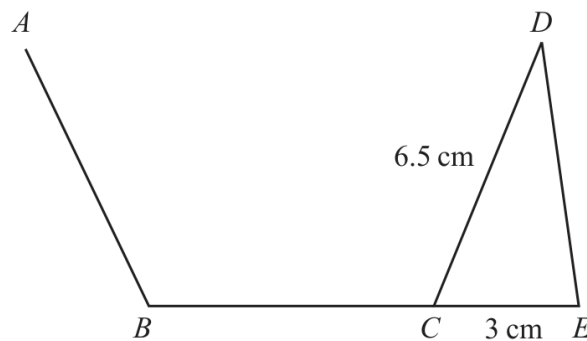
Find the size of the largest angle in triangle  $ABC$   
Give your answer correct to the nearest degree.

o

(Total for Question 25 is 6 marks)

**Q9****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q15

**CH 4 • TOPIC 11****3 marks****15**Diagram **NOT**  
accurately drawn

$AB$ ,  $BC$  and  $CD$  are three sides of a regular pentagon and  $CDE$  is a triangle.  
 $BCE$  is a straight line.

$$CD = 6.5 \text{ cm} \quad CE = 3 \text{ cm}$$

Work out the area of triangle  $CDE$

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$ **(Total for Question 15 is 3 marks)**

**Q10****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q15

CH 4 • TOPIC 11 **3 marks**

15 The diagram shows isosceles triangle  $EFG$

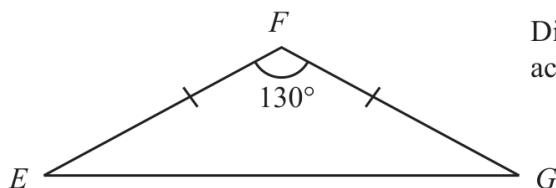


Diagram **NOT**  
accurately drawn

$$EF = GF$$

$$\text{Angle } EFG = 130^\circ$$

The area of triangle  $EFG$  is  $74 \text{ cm}^2$

Work out the length of  $EF$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 15 is 3 marks)

## 02 The Sine Rule

You need one complete pair: a side and the angle opposite it. Put the unknown on top: sides over sines to find a side, sines over sides to find an angle.

### WHEN YOU SEE → YOU DO

a side and its opposite angle + one more

→ sine rule

want a side

$$\rightarrow \frac{a}{\sin A} = \frac{b}{\sin B}$$

want an angle

$$\rightarrow \frac{\sin A}{a} = \frac{\sin B}{b}$$

“given that angle  $PQR$  is obtuse”

→  $180^\circ$  – the calculator angle

### WORKED MODEL • JUNE 2022 P1H

$AB = 9.7$ ,  $BC = 12.3$ , angle  $B = 115^\circ$ . Find angle  $C$ .

1  $AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos 115^\circ \Rightarrow AC = 18.5$

cosine rule first

2  $\sin C = \frac{9.7 \sin 115^\circ}{18.5}$

sine rule

ANSWER  $C = 28.2^\circ$

× **TRAP** The sine rule gives an acute angle. If the angle is obtuse, use  $180^\circ$  – that angle.

## IDEA 02

**Practise: The Sine Rule**

My questions first, then selected practice, then real past papers.

**Q11** DR ESLAM Sine rule

CH 4 • TOPIC 11 3 marks

In triangle  $ABC$ ,  $A = 40^\circ$ ,  $B = 75^\circ$ ,  $BC = 12$  cm. Find  $AC$ .

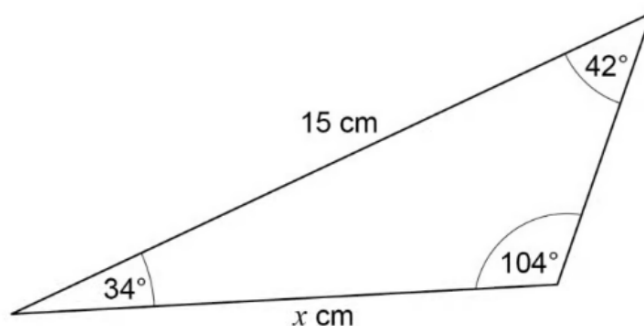
## SELECTED PRACTICE

**Q12****PRACTICE**

Practice • Medium

CH 4 • TOPIC 11 1 mark

6 Here is a triangle.



Not drawn accurately

Circle the correct equation.

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| $\frac{\sin x}{42} = \frac{\sin 15^\circ}{104}$ | $\frac{x}{\sin 42^\circ} = \frac{15}{\sin 104^\circ}$ |
| $\frac{\sin x}{34} = \frac{\sin 15^\circ}{104}$ | $\frac{x}{\sin 42^\circ} = \frac{15}{\sin 34^\circ}$  |

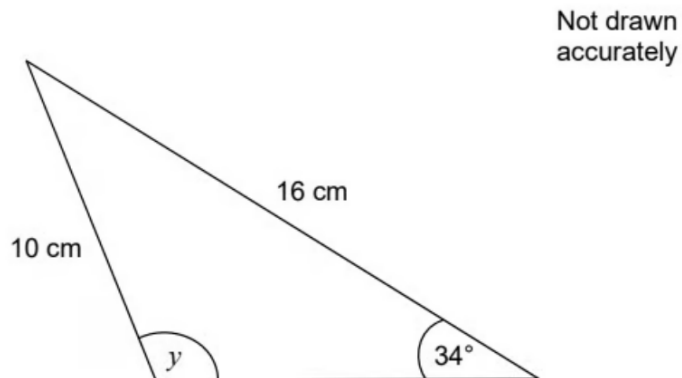
(1 mark)

**Q13****PRACTICE**

Practice • Medium

CH 4 • TOPIC 11   **3 marks**

**13** In the triangle, angle  $y$  is obtuse.



Work out the size of angle  $y$ .

.....degrees

**(3 marks)**

**Q14** PRACTICE Practice • Hard

CH 4 • TOPIC 11 5 marks

- 3 The diagram shows triangle  $LMN$ .

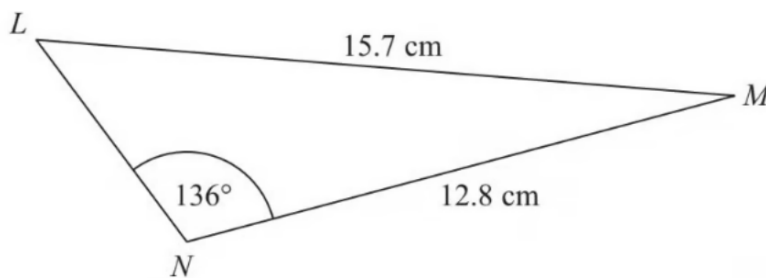


Diagram **NOT**  
accurately drawn

Calculate the length of  $LN$ .

Give your answer correct to 3 significant figures.

(5 marks)

**Q15****PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 11 4 marks

8 The diagram shows triangle  $ABC$

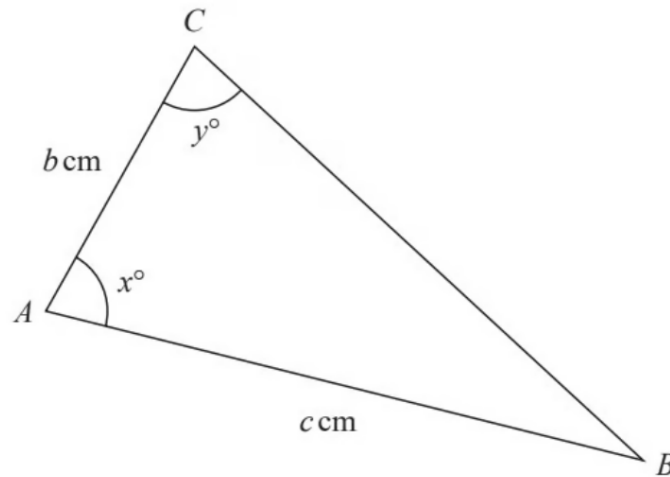


Diagram **NOT**  
accurately drawn

$c = 11.5$  correct to one decimal place

$x = 80$  correct to the nearest whole number

$y = 75$  correct to the nearest whole number

Calculate the upper bound for the value of  $b$ . Show your working clearly.

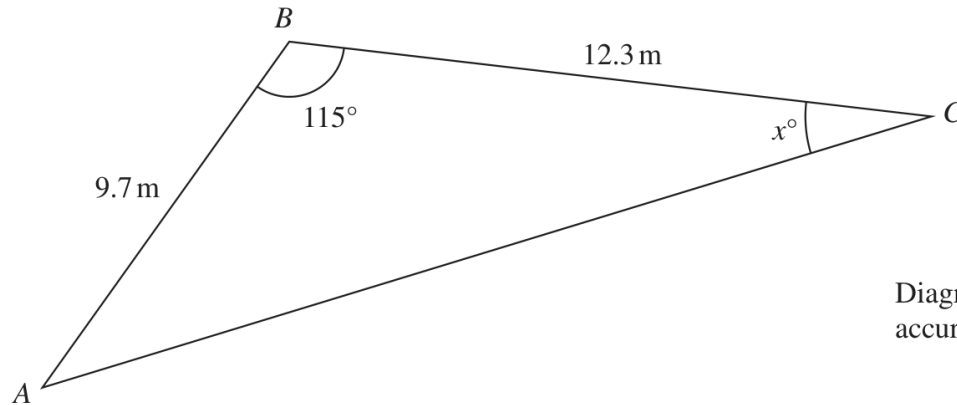
Give your answer correct to 3 significant figures.

**(4 marks)**

## PAST PAPERS

**Q16** PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q18

CH 4 • TOPIC 11 5 marks

18 Here is triangle  $ABC$ Diagram **NOT**  
accurately drawn

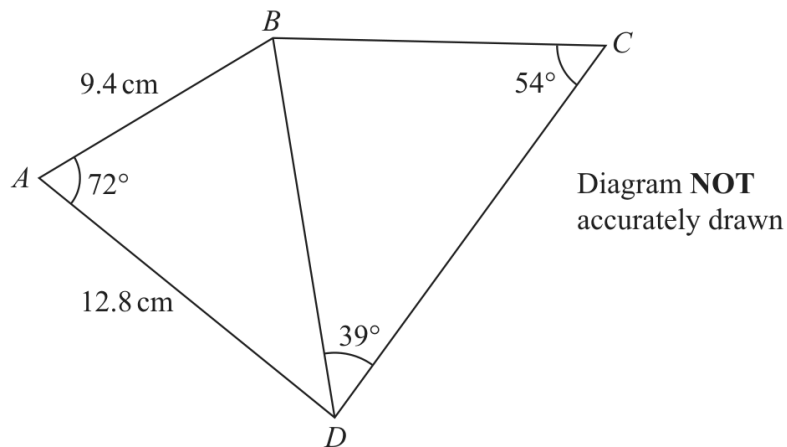
Work out the value of  $x$   
Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$ 

(Total for Question 18 is 5 marks)

**Q17****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q20

**CH 4 • TOPIC 11 5 marks****20**Work out the length of  $BC$ 

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 5 marks)**

## 03 The Cosine Rule

Third side: two sides and the angle between. An angle from three sides:  
rearrange to  $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ .

### WHEN YOU SEE → YOU DO

**two sides and the angle between**

$$\longrightarrow a^2 = b^2 + c^2 - 2bc \cos A$$

**three sides**

$$\longrightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

**“the largest angle”**

→ opposite the longest side

**bearings**

→ use the angle between the paths

### WORKED MODEL • NOVEMBER 2025 P1H

$AB = 8$ ,  $BC = 10$ ,  $CA = 9$ . Find angle  $BAC$ .

$$1 \quad \cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)} = \frac{45}{144}$$

rearranged cosine rule

**ANSWER**  $A = 71.8^\circ$

× **TRAP** Doing  $(b^2 + c^2 - 2bc) \cos A$ . Work out  $2bc \cos A$  first, then subtract.

## IDEA 03

**Practise: The Cosine Rule**

My questions first, then selected practice, then real past papers.

**Q18** DR ESLAM Cosine rule

CH 4 • TOPIC 11 3 marks

Find the third side of a triangle with sides 5 cm and 8 cm and angle  $60^\circ$  between them.

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**Q19** DR ESLAM Angle from three sides

CH 4 • TOPIC 11 3 marks

A triangle has sides 5, 7 and 9 cm. Find its largest angle.

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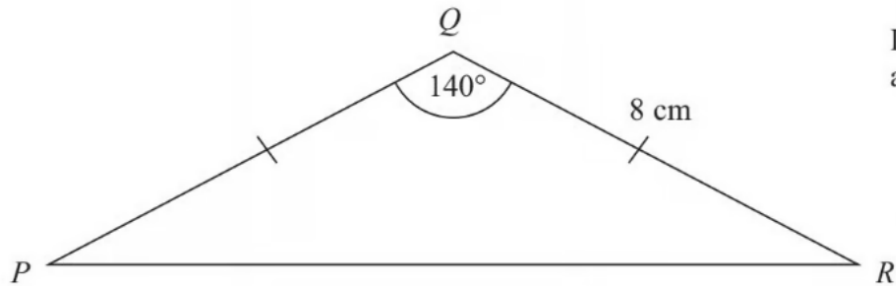
## SELECTED PRACTICE

**Q20****PRACTICE**

Practice • Medium

CH 4 • TOPIC 11 3 marks

2

Diagram **NOT**  
accurately drawn

Calculate the length of  $PR$ . Give your answer correct to 3 significant figures.

**(3 marks)**

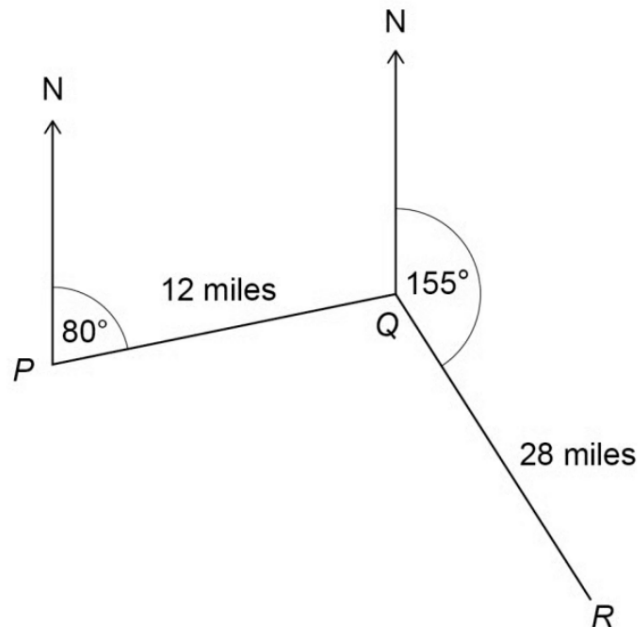
**Q21** PRACTICE Practice • Medium

CH 4 • TOPIC 11 4 marks

9 A ship sails from  $P$  to  $Q$  and then from  $Q$  to  $R$ .

$Q$  is 12 miles from  $P$ , on a bearing of  $080^\circ$

$R$  is 28 miles from  $Q$ , on a bearing of  $155^\circ$

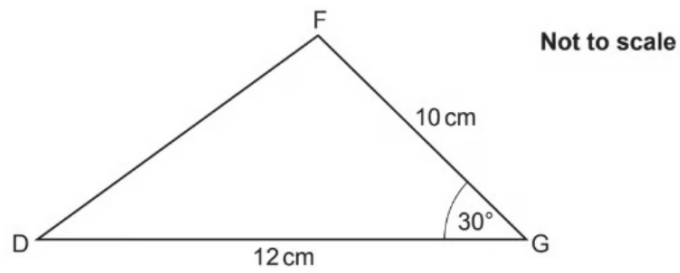


Work out the direct distance from  $P$  to  $R$ .

(4 marks)

**Q22****PRACTICE**

Practice • Medium

CH 4 • TOPIC 11 **3 marks****16** Calculate length DF in this triangle.

..... cm

**(3 marks)**

**Q23****PRACTICE**

Practice • Hard

CH 4 • TOPIC 11 5 marks

**15**  $A$ ,  $B$  and  $C$  are three towns.

The bearing of  $B$  from  $A$  is  $105^\circ$

The bearing of  $C$  from  $B$  is  $230^\circ$

The distance of  $C$  from  $A$  is 180 km.

The distance of  $C$  from  $B$  is 95 km.

Calculate the distance of  $B$  from  $A$ .

Give your answer correct to 3 significant figures.

**(5 marks)**

## PAST PAPERS

**Q24** PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1H • Q14

CH 4 • TOPIC 11 5 marks

14 The diagram shows parallelogram  $EFGH$ .

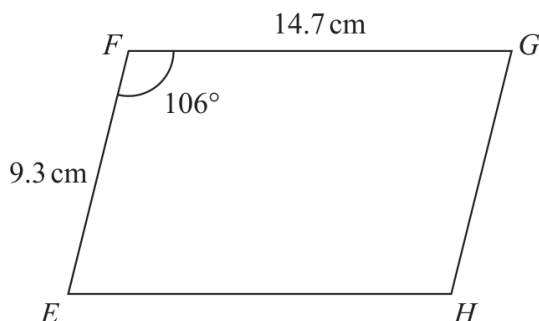


Diagram **NOT**  
accurately drawn

$EF = 9.3 \text{ cm}$   
 $FG = 14.7 \text{ cm}$   
Angle  $EFG = 106^\circ$

- (a) Work out the area of the parallelogram.  
Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$

- (b) Work out the length of the diagonal  $EG$  of the parallelogram.  
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 14 is 5 marks)

**Q25****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q19

**CH 4 • TOPIC 11 3 marks**

19 The diagram shows triangle  $ABC$

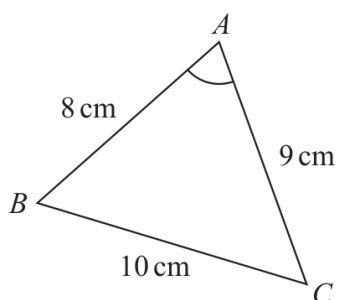


Diagram **NOT**  
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle  $BAC$

Give your answer correct to one decimal place.

o

(Total for Question 19 is 3 marks)

## 04 Multi-Step Problems

Two triangles share a side: find it in the first triangle, then use it in the second. Keep full calculator values between steps.

### WHEN YOU SEE → YOU DO

|                                     |                                             |
|-------------------------------------|---------------------------------------------|
| a quadrilateral split by a diagonal | → diagonal first                            |
| an obtuse angle hint                | → choose $180^\circ$ - the sine rule angle  |
| a flagpole on horizontal ground     | → 2D triangle first, then tan for elevation |
| area of a quadrilateral             | → add the two triangle areas                |

Pick the triangle with enough facts

Find the shared side

Second triangle

Answer

### WORKED MODEL • NOVEMBER 2021 P2H

$AB = 9$ ,  $BC = 12$ , angle  $B = 60^\circ$ ;  $CD = 7$ , angle  $ACD = 84^\circ$ . Area of  $ABCD$ ?

1  $AC^2 = 81 + 144 - 108 = 117 \Rightarrow AC = 10.82$

cosine rule

2  $\frac{1}{2}(9)(12)\sin 60^\circ + \frac{1}{2}(10.82)(7)\sin 84^\circ$

two areas

ANSWER  $46.8 + 37.6 = 84.4 \text{ cm}^2$

## IDEA 04

**Practise: Multi-Step Problems**

My questions first, then selected practice, then real past papers.

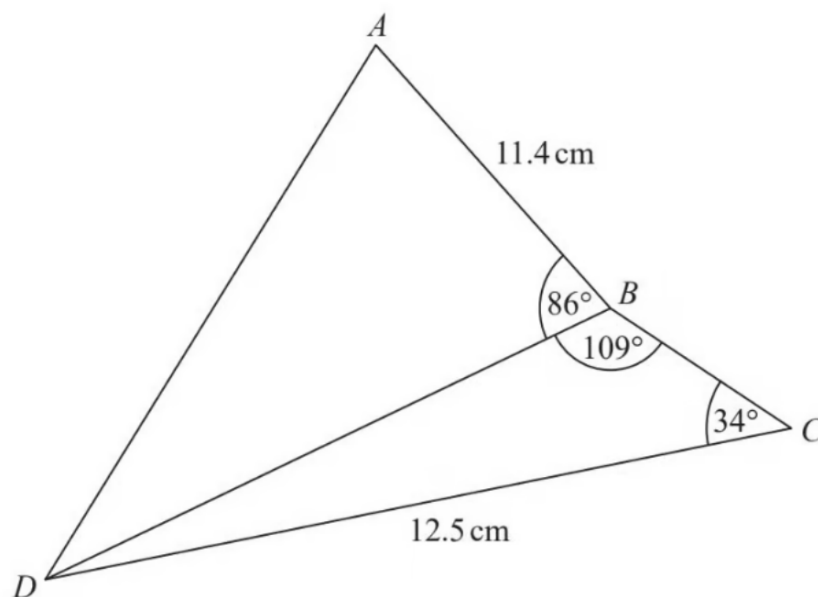
**SELECTED PRACTICE****Q26**

PRACTICE

Practice • Hard

CH 4 • TOPIC 11 5 marks

1



Work out the length of  $AD$ .

Give your answer correct to 3 significant figures.

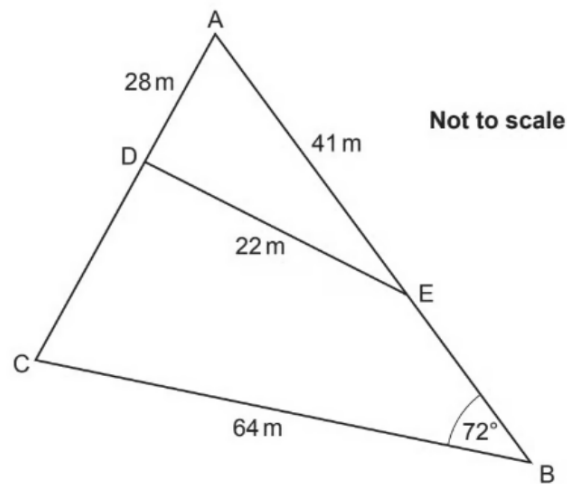
**(5 marks)**

**Q27****PRACTICE**

Practice • Hard

CH 4 • TOPIC 11 6 marks

- 13 The diagram shows triangle ABC with D on AC and E on AB. DE is a straight line.



AD = 28 m, AE = 41 m, DE = 22 m and BC = 64 m.

Calculate the length CD.

..... m

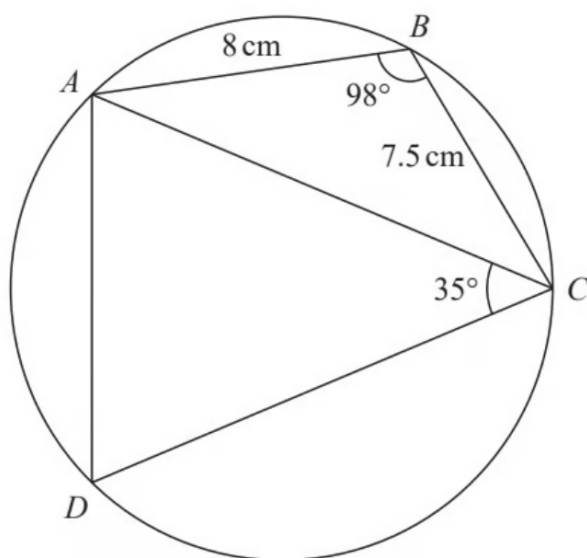
(6 marks)

**Q28****PRACTICE**

Practice • Very Hard

CH 4 • TOPIC 11 6 marks

9

Diagram **NOT**  
accurately drawn

$ABCD$  is a quadrilateral where  $A$ ,  $B$ ,  $C$  and  $D$  are points on a circle.

$$AB = 8 \text{ cm}$$

$$BC = 7.5 \text{ cm}$$

$$\text{Angle } ABC = 98^\circ$$

$$\text{Angle } ACD = 35^\circ$$

Work out the perimeter of quadrilateral  $ABCD$ .

Give your answer correct to one decimal place.

..... cm

(6 marks)

**Q29****PRACTICE**

Practice • Very Hard

CH 4 • TOPIC 11 6 marks

- 13 The diagram shows two circles with centre  $O$  and a regular pentagon  $ABCDE$

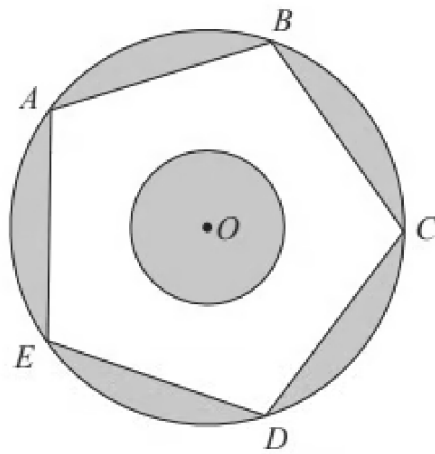


Diagram **NOT**  
accurately drawn

$A$ ,  $B$ ,  $C$ ,  $D$  and  $E$  are points on the larger circle. The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

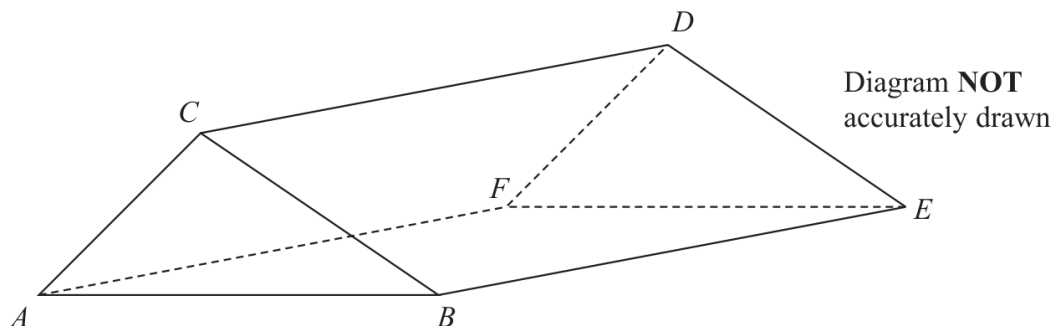
Work out the radius of the smaller circle. Give your answer correct to 3 significant figures

**(6 marks)**

## PAST PAPERS

**Q30** PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1HR • Q21

CH 4 • TOPIC 11 6 marks

21 The diagram shows the prism  $ABCDEF$  with cross section triangle  $ABC$ .Angle  $BEC = 40^\circ$  and angle  $ACB$  is obtuse. $AC = 6\text{ cm}$  and  $CE = 13\text{ cm}$ The area of triangle  $ABC$  is  $22\text{ cm}^2$ Calculate the length of  $AB$ .

Give your answer correct to one decimal place.

..... cm

(Total for Question 21 is 6 marks)

**Q31****PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q18

**CH 4 • TOPIC 11 5 marks**

18 Here is a quadrilateral  $ABCD$ .

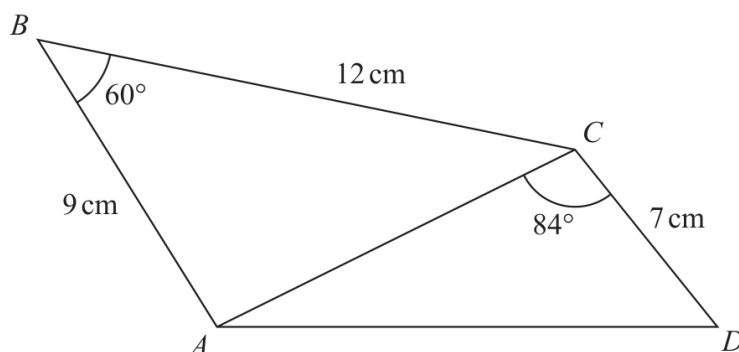


Diagram **NOT**  
accurately drawn

Calculate the area of quadrilateral  $ABCD$ .

Give your answer correct to 3 significant figures.

Show your working clearly.

.....  $\text{cm}^2$

**(Total for Question 18 is 5 marks)**

**Q32****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q16

**CH 4 • TOPIC 11 5 marks**

16 The diagram shows quadrilateral  $ABCD$

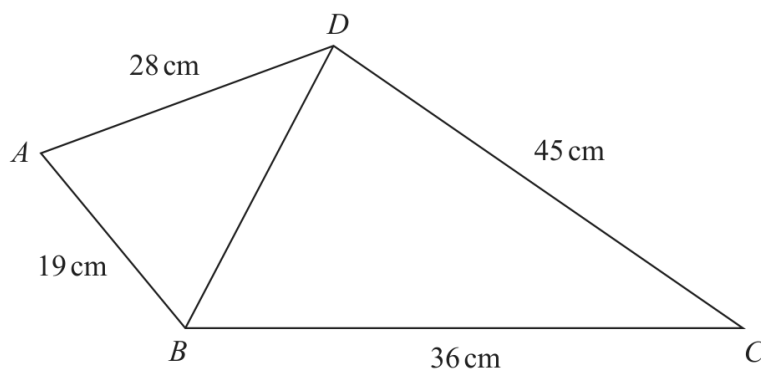


Diagram **NOT**  
accurately drawn

The angle  $BCD$  is acute.

Given that the area of triangle  $BCD = 405 \text{ cm}^2$

work out the size of angle  $ABD$

Give your answer correct to one decimal place.

○

(Total for Question 16 is 5 marks)

**Q33****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q23

**CH 4 • TOPIC 11 6 marks**

23 The diagram shows triangle  $PQR$

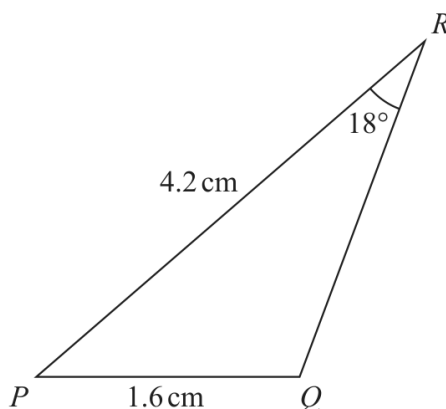


Diagram **NOT**  
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle  $PQR$  is obtuse,

work out the area of triangle  $PQR$

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$

(Total for Question 23 is 6 marks)

**Q34** PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1HR • Q17

CH 4 • TOPIC 11 5 marks

17 Here is a parallelogram  $PQRS$ , in which angle  $SPQ$  is acute.

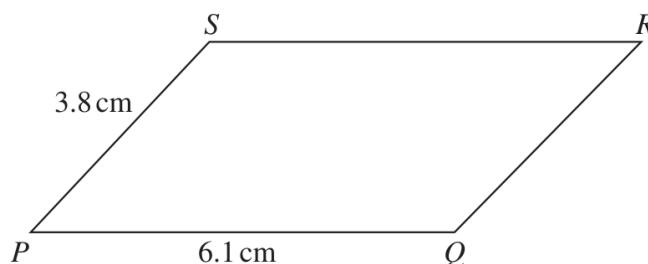


Diagram **NOT**  
accurately drawn

$$PQ = 6.1 \text{ cm} \quad PS = 3.8 \text{ cm}$$

The area of the parallelogram is  $18 \text{ cm}^2$

Work out the length of  $QS$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 17 is 5 marks)

**Q35****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q22

**CH 4 • TOPIC 11 5 marks**

22 The diagram shows a triangle  $ABC$  and a flagpole  $BF$

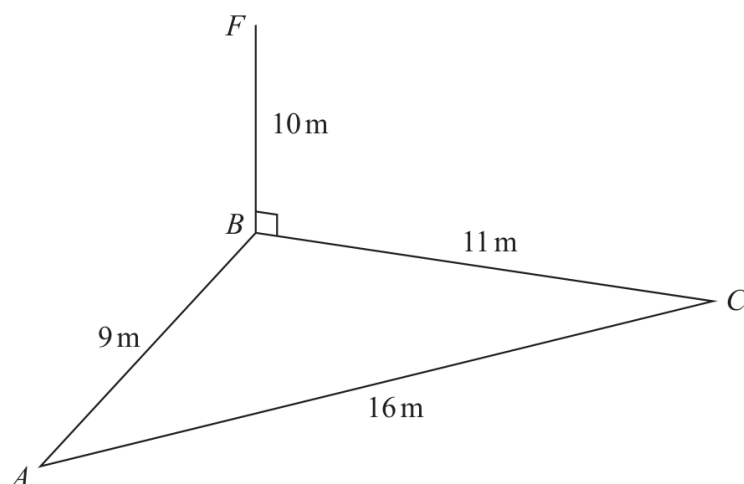


Diagram **NOT**  
accurately drawn

$A$ ,  $B$  and  $C$  are points on horizontal ground.

$BF$  is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

$D$  is the point on  $AC$  such that angle  $BDC = 90^\circ$

Work out the size of the angle of elevation of the point  $F$  from the point  $D$

Give your answer correct to one decimal place.

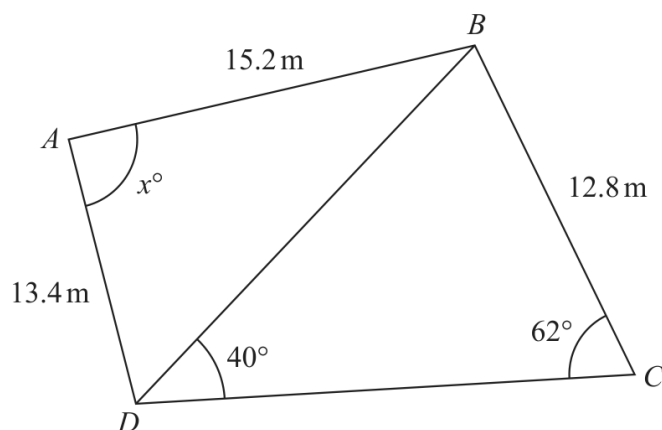
o

(Total for Question 22 is 5 marks)

**Q36****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q19

CH 4 • TOPIC 11 5 marks

19 Here is quadrilateral  $ABCD$ Diagram **NOT**  
accurately drawnWork out the value of  $x$ 

Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$ 

(Total for Question 19 is 5 marks)

# ALL The Whole Topic on One Page

Tear this page out before the exam.

- |           |                    |                                                          |
|-----------|--------------------|----------------------------------------------------------|
| <b>01</b> | <b>Area</b>        | $\frac{1}{2}ab \sin C$ with the included angle.          |
| <b>02</b> | <b>Sine rule</b>   | A matching pair; obtuse $\rightarrow 180^\circ -$ angle. |
| <b>03</b> | <b>Cosine rule</b> | Two sides + included angle, or three sides.              |
| <b>04</b> | <b>Multi-step</b>  | Shared side first; no rounding in the middle.            |

**BOTTOM LINE** Matching pair  $\rightarrow$  sine • sandwich angle or 3 sides  $\rightarrow$  cosine  
• area  $\rightarrow \frac{1}{2}ab \sin C$ .

# Answers

Sine Rule, Cosine Rule and Area of a Triangle • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • AREA $\frac{1}{2}ab \sin C$

- Q1** Dr Eslam • Area  $\frac{1}{2}(7)(9) \sin 50^\circ = 24.1 \text{ cm}^2$
- Q2** Practice • Medium (a)  $18.2 \text{ cm}^2$  (b)  $AB = 6.56 \text{ cm}$
- Q3** Practice • Hard  $BC = 8.91$ ;  $29.3 \text{ cm}^2$
- Q4** Practice • Hard (a)  $17.6 \text{ m}^2$  per side,  $105.7 \text{ m}^2$ : 13 bottles (b)  $x = 40.2^\circ$
- Q5** Practice • Very Hard  $B = 74.6^\circ$ ,  $AC = 7.83$ ;  $ACB = 41.6^\circ$
- Q6** Edexcel 4MA1 • January 2022 • Paper 1H • Q9 height 10.5:  $147 \text{ cm}^2$
- Q7** Edexcel 4MA1 • June 2022 • Paper 2HR • Q15  $2 \times \frac{1}{2}(6)(11) \sin 118^\circ = 58.3 \text{ cm}^2$
- Q8** Edexcel 4MA1 • May 2023 • Paper 1HR • Q25  $x = 3.5$ : sides 6, 8,  $BC = 4.11$ : largest angle  $103^\circ$
- Q9** Edexcel 4MA1 • June 2024 • Paper 2HR • Q15 angle  $DCE = 72^\circ$ :  $9.27 \text{ cm}^2$
- Q10** Edexcel 4MA1 • May 2024 • Paper 1HR • Q15  $\frac{1}{2}x^2 \sin 130^\circ = 74$ :  $EF = 13.9 \text{ cm}$

## IDEA 02 • THE SINE RULE

**Q11** Dr Eslam • Sine rule

$$\frac{12 \sin 75^\circ}{\sin 40^\circ} = 18.0 \text{ cm}$$

**Q12** Practice • Medium

$$\frac{x}{\sin 42^\circ} = \frac{15}{\sin 104^\circ}$$

$$y = 116.5^\circ$$

**Q13** Practice • Medium

$$LN = 3.73 \text{ cm}$$

**Q14** Practice • Hard

$$b_{\max} = \frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.25$$

**Q15** Practice • Very Hard

$$AC = 18.5: x = 28.2$$

**Q16** Edexcel 4MA1 • June 2022 • Paper 1H • Q18

$$BD = 13.34: BC = 10.4 \text{ cm}$$

**Q17** Edexcel 4MA1 • June 2025 • Paper 1H • Q20**IDEA 03 • THE COSINE RULE****Q18** Dr Eslam • Cosine rule

$$\sqrt{25 + 64 - 40} = 7 \text{ cm}$$

**Q19** Dr Eslam • Angle from three sides

$$\cos^{-1} \frac{25+49-81}{70} = 95.7^\circ$$

**Q20** Practice • Medium

$$PR = 15.0 \text{ cm}$$

**Q21** Practice • Medium

$$PR = 33.2 \text{ miles}$$

**Q22** Practice • Medium

$$DF = 6.01 \text{ cm}$$

**Q23** Practice • Hard

$$\text{angle } ABC = 55^\circ: AB = 217 \text{ km}$$

**Q24** Edexcel 4MA1 • January 2020 • Paper 1H • Q14

$$(a) 131 \text{ cm}^2 \text{ (b) } 15.1 \text{ cm}$$

**Q25** Edexcel 4MA1 • November 2025 • Paper 1H • Q19

$$71.8^\circ$$

**IDEA 04 • MULTI-STEP PROBLEMS**

- Q26** Practice · Hard  $BD = 7.39$ :  $AD = 13.1$  cm
- Q27** Practice · Hard  $A = 30.4^\circ$ ,  $AC = 120.4$ :  $CD = 92.4$  m
- Q28** Practice · Very Hard  $AC = 11.7$ : perimeter 32.8 cm
- Q29** Practice · Very Hard  $r = 3.45$  cm
- Q30** Edexcel 4MA1 · January 2020 · Paper 1HR · Q21  $BC = 8.36$ ,  $C$  obtuse:  $AB = 12.4$  cm
- Q31** Edexcel 4MA1 · November 2021 · Paper 2H · Q18  $AC = 10.8$ :  $46.8 + 37.6 = 84.4$  cm<sup>2</sup>
- Q32** Edexcel 4MA1 · January 2022 · Paper 2HR · Q16  $BD = 22.7$ :  $ABD = 83.9^\circ$
- Q33** Edexcel 4MA1 · January 2022 · Paper 2H · Q23  $Q = 163.0^\circ$  (obtuse):  $1.98$  cm<sup>2</sup>
- Q34** Edexcel 4MA1 · June 2022 · Paper 1HR · Q17  $QS = 4.74$  cm
- Q35** Edexcel 4MA1 · June 2023 · Paper 2H · Q22  $BD = 5.95$ :  $59.2^\circ$
- Q36** Edexcel 4MA1 · May 2024 · Paper 1H · Q19  $BD = 17.58$ :  $x = 75.6$